METRIC

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PERFORMANCE SPECIFICATION

COOLER-DEWAR GROUP, OR-334/V, OR-335/V

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers only the Cooler-Dewar Group to be used in the M1 Thermal Imaging System (TIS). The Cooler-Dewar Group is a component of the Thermal Imaging Systems (FLIR) used in the M1. The group consists of a 120 element infrared detector DT-617A/UA, a one watt cryogenic cooler, (HD-1033B/UA, HD-1033C/UA or HD-1033D/UA), a thermal conductor and associated fastening hardware.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 5855

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2.2 Government documents. The following drawings and purchase description form a part of this document to the extent specified. Unless otherwise specified, the issue is that cited in the solicitation (see 6.2).

DRAWINGS

54490 5008606	-	Test Fixture Assembly Microphonics, Standard
54490 5008608	-	Mounting Requirements
54490 5008612	-	Cooler-Dewar Subassembly, 120 Element

PURCHASE DESCRIPTION

PD-0182-001A(CR)	-	Cooler, Cryogenic Long Life, One Watt Integral HD-1033D/UA

(Copies of the drawings and purchase description are available from the Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, New Jersey, 07703)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.2 Item definition. The Cooler-Dewar Group is a component of the Thermal Imaging Systems (FLIR) used in the M1. The group consists of a 120 element infrared detector DT-617A/UA, a one watt cryogenic cooler, (HD-1033B/UA, HD-1033C/UA or HD-1033D/UA), a thermal conductor and associated fastening hardware.

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3.2.1 Common module selection. The detector and cooler shall meet all interface requirements of their respective specifications and be inspected in accordance with the verification provisions of these specifications prior to assembly of the Cooler-Dewar Group, OR-334/V, OR-335/V (see 6.2i).

Detector-Dewar	DT-617A/UA		MIL-D-49457
Cooler, Cryogenic Mechanical	HD-1033B/UA HD-1033C/UA	OR	MIL-C-49175
OR			
Cooler, Cryogenic Long Life, One Watt Integral	HD-1033D/UA		PD-0182-001A(CR)

3.2.2 Thermal conductor. The bellows is a separate metal piece, installed internally, that thermally connects the cooler cold finger to the detector-Dewar. The installation procedure of the thermal conductor shall be approved by the Government (see 6.2.l).

3.3 Materials and manufacturing processes. Unless otherwise specified, the materials and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the Government fully meet the performance and environmental requirements specified herein.

3.4 Performance characteristics. The Cooler-Dewar Group shall meet the following performance characteristics.

3.4.1 Detector-Dewar defects. Unless otherwise specified herein, the Detector-Dewar shall be operated at 80 Kelvin (K) (+5, -20K). The Detector-Dewar shall contain no more than 12 defective elements as defined in 6.3.2, with no more than 5 hard defects.

3.4.1.1 Central element defects. In the central 18 elements of the array there shall be no hard defects and no more than 2 soft defects. The central 36 elements shall contain no hard defects and no more than 4 soft defects.

3.4.1.2 Hard defect separation. Hard defects shall be separated from one another by at least 4 non-defective elements and shall be electrically strapped to an adjacent element with common ground.

3.4.1.3 Soft defect separation. Soft defects located within the central 36 elements shall be separated from any other defect by at least 4 non-defective channels. Soft defects shall not be strapped (see 6.3.2.2) unless the strapping is from an adjacent hard defect at the end of a group of 5 commonly grounded elements (see 6.3.5).

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3.4.2 Microphonics. When tested in accordance with paragraph 4.5.2, no elements of the detector array shall have noise peaks greater than 6dB above the white noise reference value from 300 to 10,000 Hz, and 10dB from 10,001 to 20,000 Hz.

3.4.3 Cooldown time. The cooldown time to reach a detector temperature of 80 K shall be in accordance with Figure 1.

3.4.4 Cooler input power. The total input power to the cooler shall not exceed that shown in Figure 2 after detector temperature has reached at least 80 K. The input voltage shall be 117 ± 2 volts, alternating current (VAC), 400 ± 20 Hertz (Hz) power source.

3.5 Environmental considerations.

3.5.1 Temperature extremes. The Cooler-Dewar shall not be damaged when operated to temperatures from -32°C to +60°C, and stored at temperatures from -54°C to +71°C (see 6.3.1.).

3.5.2 Vibration exposure. The Cooler-Dewar Group shall not be damaged by vibration to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three (3) mutually perpendicular axes.

Random:	1×10^{-3} g^2 / Hz, 5 to 500 Hz
Sine:	0.26 g's, Octave 1 42 to 84 Hz X axis (cooler piston axis)
	0.78 g's, Octave 1 42 to 84 Hz Y axis (cooler motor axis)
	0.85 g's, Octave 1 42 to 84 Hz Z axis (cooler coldfinger axis)

3.5.3 Temperature shock. The Cooler-Dewar shall not be damaged (see 6.3.1); by sudden changes in temperature between -54°C and +71°C.

3.6 Workmanship. All components and parts shall be manufactured and assembled to ensure uniform conformance and be free from defects.

4. VERIFICATION

4.1 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 First article inspection. When specified in the contract, the first article inspection shall be performed by the contractor. Reliability testing is required only on the cooler for first article testing.

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4.2.1 Static inspections. Equipment shall be inspected for missing or damaged parts or components. All materials, parts, processes and assemblies shall be examined for conformance to acceptable workmanship for electronic equipment. For additional guidance information see 6.5.

4.2.1.1 Cleaning. After fabrication, parts and assembled equipment shall be cleaned of smudges; loose, spattered, or excess solder; weld metal; metal chips and mold release agents; or any other foreign material which might detract from the intended operation, function, or appearance of the equipment.

4.2.1.2 Threaded parts or devices. Screws, nuts and bolts shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs, and shall be firmly secured.

4.2.1.3 Wiring. Wires and cables shall be positioned or protected to avoid contact with rough or irregular surfaces and sharp edges and to avoid damage to conductors or adjacent parts.

4.2.1.4 Shielding. Shielding on wires and cables shall be secured in a manner that will prevent it from contacting or shorting exposed current carrying parts. The ends of the shielding or braid shall be secured to prevent fraying.

4.2.2 Dynamic inspections. Upon successful completion of the inspections specified in 4.3.1, four first article assemblies shall be subjected to all dynamic inspections listed in Table I. The four first article coolers must successfully meet the requirement of the respective cooler specification. Tests may be conducted in any order. Failure of any tests shall constitute first article failure.

4.2.3 Disposition of first article samples. Cooler-Dewar Groups which have been subjected to and passed first article inspections may be accepted on the contract or purchase order; provided all visible and known damage is repaired and the assemblies are re-subjected to and pass Group A inspections. The four reliability first article coolers shall not be considered as part of the procurement quantities.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Defective elements	3.4.1	4.5.1
Detector microphonics	3.4.2	4.5.2
Cooldown time	3.4.3	4.5.3
Cooler input power	3.4.4	4.5.4
Environmental considerations	3.5	4.6

4.3 Inspection procedures for verification. Unless otherwise specified herein, the Cooler-Dewar-Group shall meet specified performance requirements when operated at an ambient temperature of +23°C ±5°C with detector temperature of 80 K +5 K, -20 K, and detector bias of +5.0 Vdc ± 0.2 Vdc and cooler power of 117 ± 2 Vac, 400 Hz ± 20 Hz.

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4.4 Conformance inspection. Every item requirement contained in Section 3 of this specification has a conformance test or inspection identified in paragraph 4.5. The subparagraph headings are traceable to the paragraph/subparagraph headings contained in Section 3.

4.4.1 Inspection. Inspection shall be as specified in 4.2.1.

4.4.2 Inspections.

4.4.2.1 Group A inspection. Group A inspection shall be conducted on all Cooler-Dewar Groups. Group A inspections listed in Table II may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Defective elements	3.4.1	4.5.1
Detector microphonics	3.4.2	4.5.2
Cooldown time	3.4.3	4.5.3
Cooler input power	3.4.4	4.5.4

4.4.2.2 Group B inspection. Group B inspection is not required.

4.4.2.3 Group C inspection. Group C inspection shall be conducted on Cooler-Dewar Groups selected from units which have passed tests in 4.4.2.1. The environmental tests of 4.6 shall be performed on a sample quantity defined in 4.4.2.3.1.

4.4.2.3.1 Sampling for group C inspection. Unless otherwise specified, the sample quantity shall be as defined below. The samples shall be randomly selected from units produced during the sampling period.

Monthly lot size	Number of units monthly
1-9	1
10-29	2
30-UP	3

4.4.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2g.).

4.4.2.3.3 Reinspection of conforming Group C samples. Unless otherwise specified, sample units which have been subjected to and passed Group C inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are retested and pass Group A inspections.

4.4.2.4 Group D inspection. Group D inspection not required.

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4.5 Performance characteristics. Unless otherwise specified, the Cooler-Dewar Group shall be interconnected and operated under standard ambient conditions (temperatures of 23°C ± 5°C and humidity not to exceed 90 percent). The detector shall be operated at +5.0 Vdc ± 0.2 Vdc. Each detector element shall be biased with current flow in the same direction. The cooler shall operate from a 117 Vac ± 2 Vac, 400 Hz ± 20 Hz power source.

4.5.1 Detector-dewar defective elements. The detector array shall be checked for continuity using an ohm meter (current limited to 1 mA). Lack of continuity indicates an open element and shall constitute failure of the tests. The Group A inspection test report on the detector shall be used to verify other requirements of paragraph 3.4.1.

4.5.2 Detector microphonics. The Cooler-Dewar Group shall be mounted to a microphonics test fixture. The cooler shall be activated to run through out the duration of the microphonics test procedure. Any detector element which fails to meet the requirements of paragraph 3.4.2 shall be considered a failure of this test.

4.5.2.1 Detector bias. Bias the detector by operating the detector resistor bias pack at +5.0, ± 0.2 Vdc. Each detector element shall be biased with current flow in the same direction.

4.5.2.2 Frequency spectral density test. When the detector array reaches an operational temperature between 60 K and 80 K, the frequency spectral density of each detector element shall be measured using a noise power spectrum analyzer. The spectrum analyzer and any associated electronics/software shall be configured and operated to provide a frequency span of 0 - 20,000 Hz, a measurement window (filter) in the Flat Top configuration and a Noise Equivalent Bandwidth of 360 Hz ± 10 Hz. A minimum of 64 samples of the frequency spectrum shall be taken and averaged.

4.5.2.3 White noise reference level. The white noise reference level of a given detector element shall be determined by averaging the noise on that element from 12,000 to 20,000 Hz with the cooler power applied.

4.5.3 Cooldown time. The required operating temperature shall be determined by measuring the voltage across the temperature sensor #1 installed in the Detector-Dewar with a 1 mA ± 0.1 mA bias applied. The Detector-Dewar shall be operated with bias voltage applied to all sample elements from when the cooler is turned on to when the above measurements are made. Failure to meet the requirements of paragraph 3.4.3 shall constitute failure of this test.

4.5.4 Cooler input power. Input power shall be measured using an AC wattmeter when the temperature sensor is equal to or less than 80 K Vdc. Failure to meet the requirements of 3.4.4 shall constitute failure of this test.

4.6 Environmental tests. Complete the environmental tests under paragraph 4.6 in the order given. MIL-STD-810E, Environmental Test Methods is cited for guidance. All test results must be documented (see 6.6).

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4.6.1 Pretest standard ambient checkout. All test items require a pretest checkout at standard ambient conditions so that baseline data can be established. Use the following guidelines:

a. Stabilize the test item at standard ambient conditions (+23°C ± 5°C).

b. Conduct a complete visual examination.

c. Document the results.

d. Conduct an operational checkout by performing the tests contained in paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4.

e. Record the results. If the test item performs satisfactorily, proceed with the temperature shock test.

4.6.2 High temperature. Subject a Cooler-Dewar Group to high temperatures to determine of the equipment can be stored and operated under hot climatic conditions without experiencing physical damage or deterioration in performance.

4.6.2.1 Storage high temperature test. Place the Cooler-Dewar Group into the test chamber. Adjust the air temperature of the test chamber to +71°C (humidity not to exceed 90%) and maintain until the temperature of the Cooler-Dewar Group stabilizes. Adjust the air temperature of the test chamber to ambient temperature (+23°C ± 5°C) and maintain until the temperature of the Cooler-Dewar Group stabilizes. Cycle the air temperature six more times between +71°C and ambient. After the last cycle, perform tests given in paragraphs 4.5.1, 4.5.3, and 4.5.4 before subjecting the Cooler-Dewar Group to High Temperature Operation testing (see 4.6.2.2). Failure to meet the requirements of these tests constitutes failure of the high temperature test.

4.6.2.2 Operation high temperature test. Adjust the air temperature of the test chamber to +60°C (humidity not to exceed 90%) and maintain until the temperature of the Cooler-Dewar Group stabilizes. Perform tests given in paragraphs 4.5.1, 4.5.3, and 4.5.4. Failure to meet the requirements of these tests constitutes failure of the high temperature test.

4.6.2.3 High temperature pass/fail criteria. Adjust the air temperature of the test chamber to ambient temperature (+23°C ± 5°C) and maintain until the temperature of the Cooler-Dewar Group stabilizes. Perform tests given in paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to meet the requirements of these tests constitutes failure of the high temperature test.

4.6.3 Low temperature. Subject a Cooler-Dewar Group to low temperatures which to determine if the equipment can be stored under low temperature conditions without experiencing physical damage or deterioration in performance. Establish a test baseline by perform the steps given in paragraph 4.6.1 before beginning the vibration test.

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4.6.3.1 Storage low temperature test. Adjust the air temperature of the test chamber to -54°C and maintain until the temperature of the Cooler-Dewar Group stabilizes. Hold the air temperature at -54°C for 24 hours.

4.6.3.2 Low temperature pass/fail criteria. Adjust the air temperature of the test chamber to ambient temperature (+23°C ± 5°C) and maintain until the temperature of the Cooler-Dewar Group stabilizes. Perform tests given in paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to meet the requirements of these tests constitutes failure of the high temperature test.

4.6.4 Vibration. Subject a Cooler-Dewar Group to vibration test which determines the resistance of the equipment to vibrational stresses expected during shipment and operational environment. Establish a test baseline by performing the steps given in paragraph 4.6.1 before beginning the vibration test.

4.6.4.1 Test parameters. Mount the Cooler-Dewar Group on a vibration table. With the Cooler-Dewar Group operating vibrate in each of the three (3) mutually perpendicular axes to the following random and superimposed sinusoidal levels for a period of 15 minutes in each axis. The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Paragraph 4.5.3 shall be accomplished during each axis.

a. Random: 1×10^{-3} g^2/Hz, 5 to 500 Hz.

b. X axis (cooler piston axis) Sine: 0.26 g’s, Octave 1 42 to 84 Hz.

c. Y axis (cooler motor axis) Sine: 0.78 g’s, Octave 1 42 to 84 Hz.

d. Z axis (cooler coldfinger axis) Sine: 0.85 g’s, Octave 1 42 to 84 Hz.

4.6.4.2 Vibration pass/fail criteria. Perform tests given in paragraph 4.5.3 during the sweep of each axis. After the vibration test is completed, perform tests given in paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to meet the requirements of paragraph 4.5.3 during vibration or paragraphs 4.5.1, 4.5.2, 4.5.3, or 4.5.4 after vibration constitutes failure of the vibration test.

4.6.5 Temperature shock. Subject the Cooler-Dewar Group to temperature shock tests to determine if the materiel can withstand sudden changes in the temperature of the surrounding atmosphere without experiencing physical damage or deterioration in performance.

4.6.5.1 Low temperature test. Place the Cooler-Dewar Group into the test chamber. Adjust the air temperature of the test chamber to -54°C and maintain for one hour or until the temperature of the Cooler-Dewar Group stabilizes whichever is longer.

4.6.5.2 High temperature transfer. Transfer the Cooler-Dewar Group to a +74°C environment in no more than five (5) minutes. Cycle the test chamber until the Cooler-Dewar Group temperature has stabilized at the high temperature.

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4.6.5.3 Low temperature transfer. Transfer the Cooler-Dewar Group to the low temperature environment (-54°C). Maintain the temperature until the Cooler-Dewar Group temperature has stabilized.

4.6.5.4 Test repetitions. Transfer the Cooler-Dewar Group in the following cycles, ensuring that the test item stabilizes at each temperature before continuing.

a. Repeat high temperature (paragraph 4.6.5.2).

b. Repeat low temperature (paragraph 4.6.5.3).

c. Repeat high temperature (paragraph 4.6.5.2).

d. Repeat low temperature (paragraph 4.6.5.3).

4.6.5.4 Temperature shock pass/fail criteria. Return the Cooler-Dewar Group to controlled ambient temperature of +23°C ± 5°C. Maintain the ambient temperature until the test unit stabilizes. Perform tests given in paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to meet the requirements of these tests constitutes failure of the temperature shock test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Cooler-Dewar Group is intended for use in the M1 Thermal Imaging System.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Issue of DoDISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.2)

c. Packaging requirements (see 5.1).

d. The Government documents listed in paragraph 2.2 are required for the performance of this contract or delivery order.

e. Necessary actions by the contractor in the event of a Group C failure.

f. If rough handling and functional tests are required.

g. Whether first article requirements for the Cooler-Dewar Group are required for groups bought under this specification.

h. Production delivery schedule.

(1) Defined in terms or monthly lots.

(2) Actions required relative to Group C failures (see 4.4.2.3.2)

(3) Disposition/refurbishment of Group C.

i. The Contractor must order Group A detector inspection reports for use in validation of the requirements of paragraphs 3.4.1 and 4.6.1; Defective Elements. The Contractor must also validate conformance with Group A, D and D inspections for both the Detector-Dewar and the Cooler, Cryogenic, Long Life or Cooler, Cryogenic, Mechanical per paragraph 3.2.1, Common Module selection.

j. Specify drawing 5008612-1 for HD-1033B/UA, HD-1033C/UA Cooler or 5008612-2 for HD-1033D/UA Cooler.

k. When first article rough handling tests are not required.

l. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packaging.

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6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, failure of any finish, hardware, connection or components, or any deterioration in quality or failure to operate within the specification requirements.

6.3.2 Defective element. A defective element should be defined as one which has failed to meet the requirements of MIL-PRF-49457B(CR). A defective element may be classified as either hard or soft.

6.3.2.1 Hard defect. A hard defect is an element which fails the requirements of MIL-PRF-49457B(CR) or which has a 500 K blackbody specific detectivity less than 6.0 x 10^9 cm $Hz^{1/2}W^{-1}$. Two elements shorted together are considered one hard defect.

6.3.2.2 Soft defect. A soft defect is one which meets all requirements of MIL-PRF-49457B(CR) and has a 500 K blackbody specific detectivity equal to or greater than 6.0 x 10^9 cm $Hz^{1/2}W^{-1}$, but less than 8,5 x 10^9 cm $Hz^{1/2}W^{-1}$.

6.3.3. Element. An element is defined as one channel in the 120 element detector array.

6.3.4 Inspection lot size. An inspection lot is defined as three months continuous production or 120 units, whichever comes first, of those assemblies which have successfully passed Group A inspection.

6.3.5 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, are considered as a single defect.

6.4 Environmental pollution preventive. Disposal methods of packaging materials, to prevent environmental pollution, are contained in MIL-HDBK-742, the material specification, or may be obtained from the preparing activity of the applicable material specification.

6.5 Workmanship guidance information.

6.5.1 Containment. The harness and cable form containment means should be neat in appearance, uniformly applied, and positioned to retain critical form factors and breakout locations. The containment means (lacing, ties, tiedown straps, etc.) should not cause the wire or cable insulation to deform so that performance characteristics are adversely affected.

6.5.2 Insulation. There should be no evidence of burns, abrading, or pinch marks in the insulation that could cause short circuits or leakage.

6.5.3 Clearance. The clearance between wires or cables and heat generating parts should be sufficient to minimize deterioration of the wires or cables.

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6.6 Environmental test guidance. MIL-STD-810E, Environmental Test Methods, is referenced as guidance for performing the environmental testing given in paragraph 4.7. The following test methods described in MIL-STD-810E apply:

a. High temperature test. Use Method 501.3, High Temperature as guidance.

b. Low temperature test. Use Method 502.3, Low Temperature as guidance.

c. Vibration test. Use Method 514.4, Vibration as guidance.

d. Temperature shock test. Use Method 503.3, Temperature Shock as guidance.

6.6.1 Test plan. A test plan should be developed before beginning environmental testing. The test plan should provide the step-by-step procedures for accomplishing the specified environmental test.

6.6.2 Test setup parameters.

a. Test facilities. One test chamber or cabinet is required to perform the high temperature test and the low temperature test. Two test chambers or cabinets are required to perform the temperature shock test. When performing the temperature shock test, unless otherwise specified, the chambers must be equipped so that, after transfer of the test item, the test conditions within the chamber can be established with five (5) minutes. Material handling equipment may be necessary for transfer of the test item between test chambers. The test chambers must be equipped with auxiliary instrumentation capable of maintain and continuously monitoring the test conditions through the envelope of air surrounding the test item.

b. Temperature. Unless otherwise specified, if any action other than test item transfer results in a temperature change of more than 2°C (3.6°F) of the test item temperature, the test item will be stabilized at the required temperature before continuing.

c. Air velocity. Air velocity in the vicinity of the test item should not exceed 1.7 m/s (325 ft/min) to provide standard testing conditions.

d. Rate of temperature change. Unless otherwise specified, the rate of temperature change should not exceed 3°C (6°F) per minute.

e. Temperature measurement. Temperature sensor instrumentation is required on or in the test item to establish temperature stabilization data.

f. Temperature recording. Continuous recordings of the chamber and test item temperature should be made if required.

g. Transfer time. During temperature shock test, transfer time between the two environments should not exceed five (5) minutes.

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6.6.3 Test interruption.

a. Undertest interruption. If the test procedure is interrupted due to work schedules, etc., the test item can be left at the test temperature or returned to the standard ambient temperature (+23°C ± 5°C) conditions for the time required. Before continuing the test, the Cooler-Dewar Group must be restabilized at the temperature of the last successfully completed period before the interruption. If, before the temperature change, the test must be reinitiated at the point of interruption and the Cooler-Dewar Group reestablished at the test condition. If the interruption occurs during the transfer, the Cooler-Dewar Group must be reestablished at the previous temperature and then transferred.

b. Overtest interruption. Any interruption that results in more extreme exposure of the test item than required, should be followed by a complete physical examination and operational checkout of the test item before continuing with testing. If a problem is discovered, it is preferable to stop the test and start over with a new test item. Not doing this, and a failure occurs during the remainder of the test, the test results could be invalid due to the overtest condition. If no problem is discovered, reestablish preinterruption conditions and continue from the point where the test tolerances were exceeded.

6.6.4 Test results documentation. The following guidelines should be followed when establishing the format and subject matter to be included in the test results documentation. Test results documentation should be prepared for each environmental test to which the item is subjected. The contractor should make this documentation a matter of permanent record and submit to the Government if requested.

a. Test item identification (manufacturer, serial number, etc.).

b. Previous test methods to which the test item has been subjected.

c. Results of each performance check, visual examination (and photographs, if applicable), and comparison with failure criteria.

(1) Pretest.

(2) During test.

(3) Post-test.

d. Length of time required for each performance check.

e. Status of the test item for each visual examination.

f. Defects noted during visual examinations.

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g. Clothing and special equipment used to set up or disassemble the test item.

h. Test temperatures.

i. Duration of each exposure.

j. Appropriate anthropometric measurements of personnel performing manipulation tests.

k. Temperature-time-versus data (test item and chamber).

l. Initial analysis of any failure.

6.7 Subject term (key word) listing.

Bellows
Cooler
Dewar

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army-CR

Preparing Activity
Army-CR
(Project 5855-0157)

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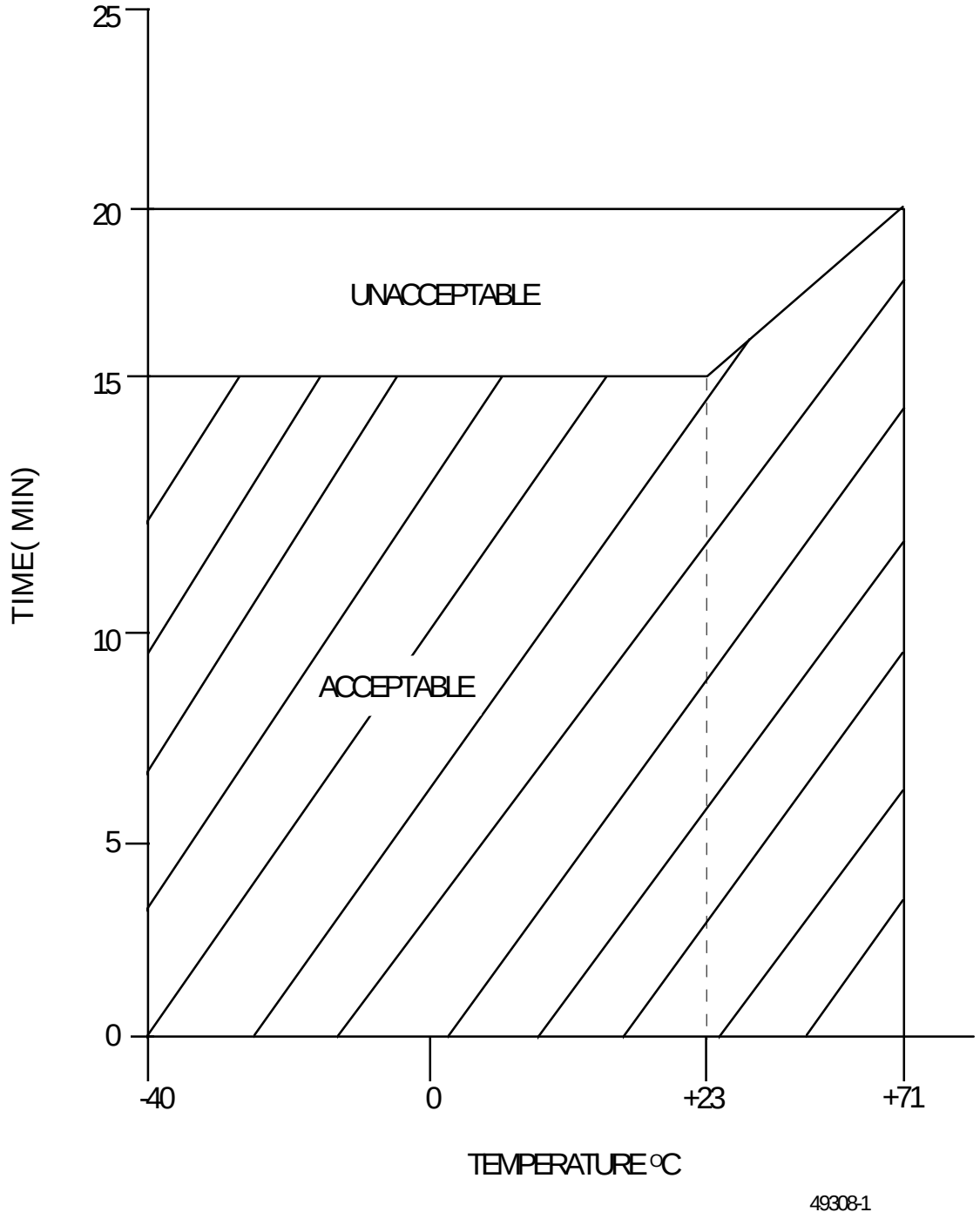


FIGURE 1.Cool down curve.

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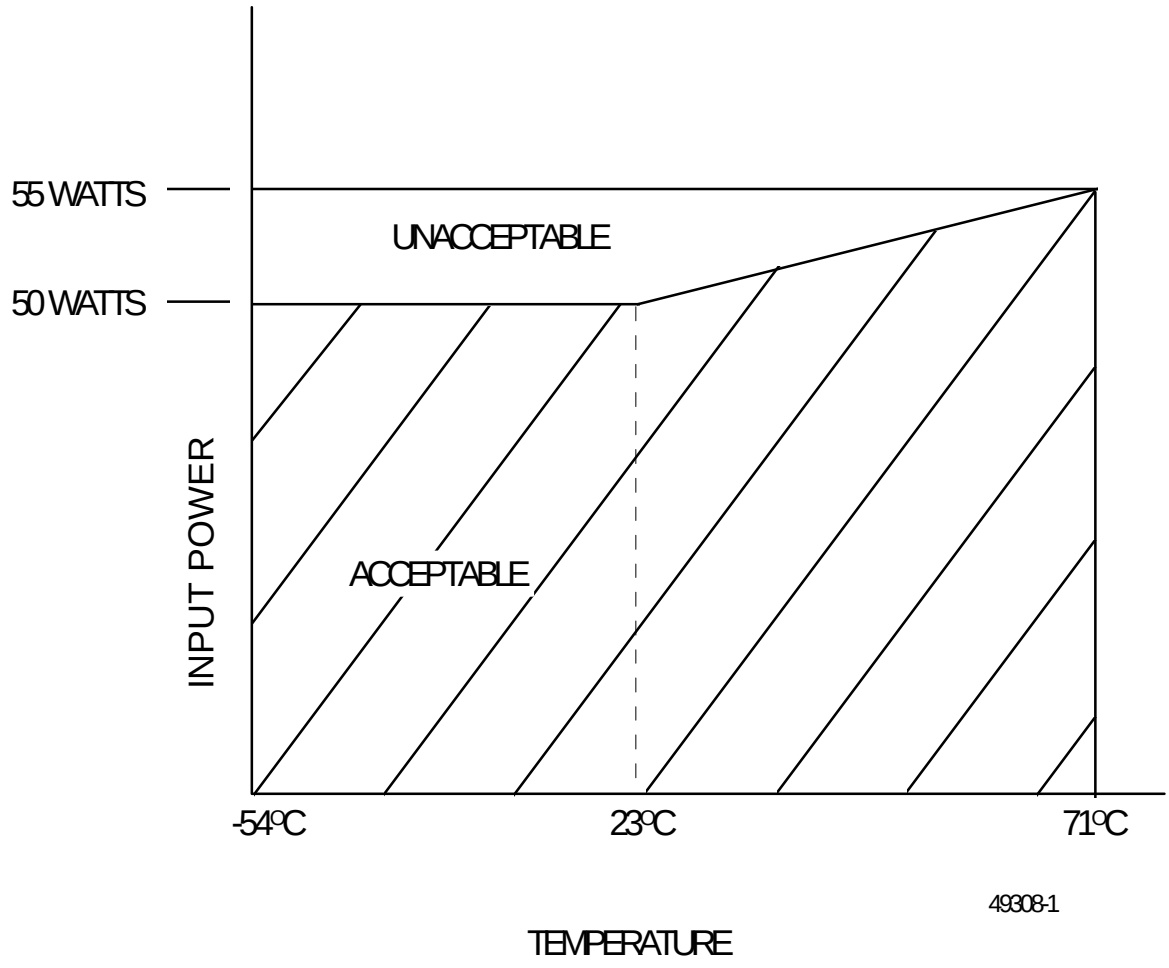


FIGURE 2.Input power with heatload applied.

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DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49308A(CR)	**2. DOCUMENT DATE (YYMMDD)** 970703

3. DOCUMENT TITLE **COOLER-DEWAR GROUP, OR-334/V, OR-335/V**

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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